# Index-calculus leads after the closure-slope study (2026-10-18)

Companion to `RESEARCH_IC_LEADS_20261018.md` in this directory. **Class:** analysis and one exploratory pilot; no `S`, no rho ratio, no claim about a deployed curve. Prior-art verdicts cite the repository files named in the note.

## The filter

Five closure arguments are settled enough that a lead failing one is not a lead: (1) the generic-group bound on table oracles, with the preprocessing-rho control measured positive on 2026-10-14 and 2026-10-16; (2) no Frobenius-stable subspace at $n = 131, 163$ (2 is a primitive root); (3) shape closure at $m \geq 3$ (every shape equals random, G1 killed); (4) the admissible-degree inequality (degree 3–5 at $w = 1$, 2 at $w = 2.2$ for $n = 131$); (5) the measured degree wall (closure 4 at $m \cdot \ell \leq 18$, above 4 at 21, degree 5 at 40 unknowns beyond a 15 GB host).

Lead map: each lead against the five closure arguments it must survive (from RESEARCH_IC_LEADS_20261018.md)

| | generic-group bound (table oracles) | no stable subspace at n = 131, 163 | shape closure at m ≥ 3 | admissible-degree inequality | measured degree wall (this host) |
|---|---|---|---|---|---|
| L1 target-free algebraic relations | pass | pass | closed: no lower degree | must still pass | closed: +1 degree (S4) |
| L2 dense-bit-row XL at degree 5 | n/a | n/a | n/a | n/a | aimed at it |
| L3 raised-cap sparse closure | n/a | n/a | n/a | n/a | closed: OOM at 7.2 GB |
| L4 4E-class stratification | pass | pass | closed: free signs | n/a | n/a |
| L5 coset-indicator degree (pilot) | pass | pass | closed: ≈ ℓ−1 on every V | n/a | n/a |
| L6 theorem: shape closure m ≥ 3 | n/a | n/a | formalises it | n/a | n/a |
| L7 lemma: admissible degree | n/a | n/a | n/a | formalises it | n/a |
| L8 K-233 derived row | pass | stable 29-blocks exist | pass | D ≤ 3–4 (cols^2.2) / 7–8 (w = 1) | n/a |
| L9 G4 stable non-subspace base | pass | escapes (not a subspace) | untested | 3n unknowns: fails | n/a |
| L10 ladder: retire online rows (Lead A) | is the bound | n/a | n/a | n/a | n/a |

Figure 1: Each lead against the five arguments. Green: the lead escapes or formalises the argument; yellow: untested or in flight; red: closed by this round.

## The leads, ranked

- **L1 target-free algebraic relations** (closed 2026-10-18): solve $S_4(x_1, ..., x_4) = 0$ with every variable in $V$ and no target. Measured in two preregistered studies. In the chained presentation, the F4 degree equals the target system at $(16, 4)$, with about 10 times the columns. In the eliminated $S_4$, the target-free system is one degree **higher** than the $S_4$ target system on 15 of 15 readings at $(13, 3)$ and $(16–19, 4)$, with no Boolean top-form dependency on the progression base at $\ell = 4$. Registered verdict: killed (`research/target_free_s4_20261018/`).
- **L2 dense XL at degree 5** (measured 2026-10-18): unmeasured on all 8 registered draws at $m \cdot \ell = 21$. The real matrix is $218,918 \times 619,790$ at $N = 40$ (17 GB dense), not the estimated 34,482 rows; every run stopped at the 6.3 GB pivot cap. Needs a 32 GB host or a sparse rank test (`research/dreg_xl_degree5_20261018/`).
- **L3 raised-cap sparse closure**: run at 2.5 G entries and killed by the memory cgroup at 7.2 GB after two hours; unmeasured. The existing dense engine scanning to degree 5 was killed twice as well (9.8 GB; allocation failure under 13.5 GB). Degree 5 at 40 unknowns needs more than this host for every engine tried.
- **L4 4E-class stratification**: closed by a sign argument; with free summand signs the $E/4E$ class never obstructs a target unless the whole base lies in $2E$ without class-2 points.
- **L5 coset-indicator degree** — piloted, negative as a predictor (below).
- **L6, L7 theorem and lemma**: write the $m \geq 3$ shape-closure argument and the admissible-degree inequality as statements the FFD programme can cite; neither exists in writing.
- **L8 K-233 derived row**: $\mathrm{ord}_{233}(2) = 29$, eight stable 29-blocks, admissible degree 3–4 under the $C(N, \leq D)^{2.2}$ cost and 7–8 under the $w = 1$ lower bound; a distance row for the one standard Koblitz curve where Frobenius structure helps the algebra. Not analysed anywhere.
- **L9 G4 stable non-subspace bases**: closed by counting (87 conditions, 22 cubic, $3n$ unknowns).
- **L10 ladder accounting**: Lead A ran positive twice; two documents still list it as scheduled.

## Pilot: the coset-indicator degree (negative)

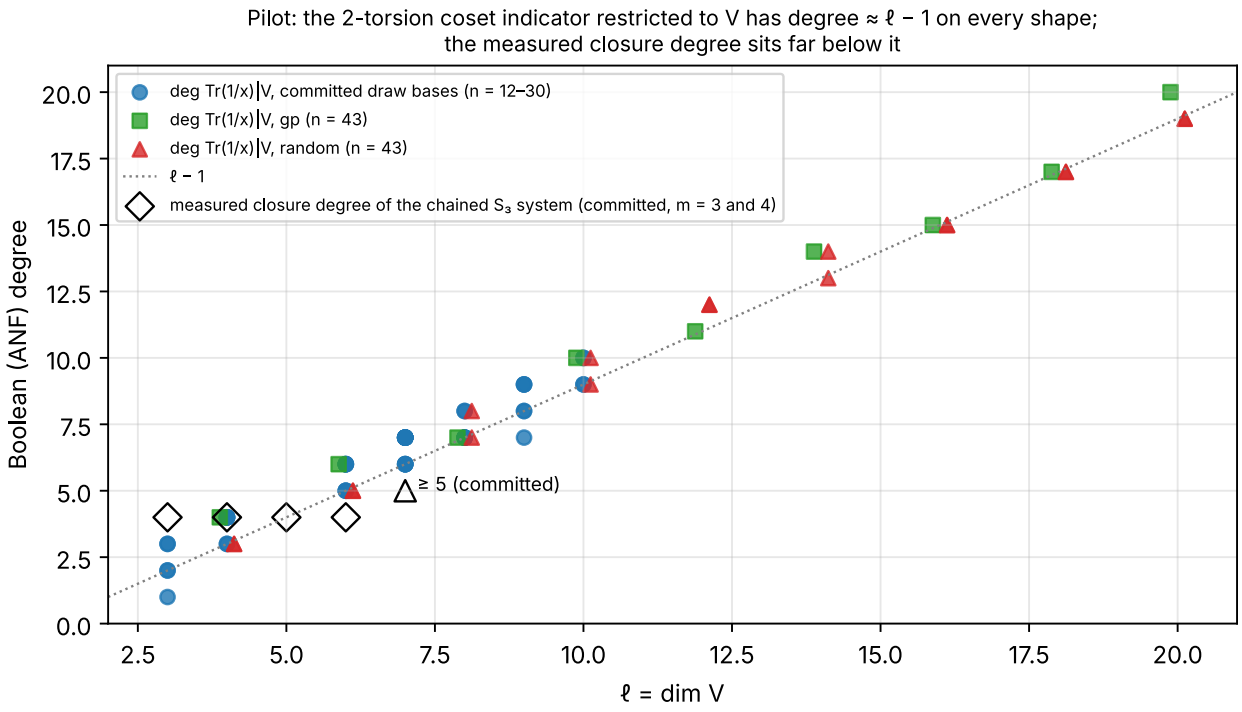



Figure 2: ANF degree of $x \to \mathrm{Tr}(1/x)$ on $V$, from `coset_indicator_pilot.jsonl` (80 records; committed anchor and fresh bases at $n = 12$–$30$, progression and random bases at $n = 43$), against the measured closure degrees of the chained system. Layout only; the degrees are exact Möbius transforms.

The indicator degree is $\ell - 1$ or $\ell$ on every base, progression no better than random, far above the closure degree 4 at $\ell \leq 6$: the closure does not refute through the parity, the indicator does not predict it, and no subspace shape at prime $n$ lowers the parity degree (Mattarei: inverse-closed subspaces are subfields). Exploratory, not preregistered.

## What would change the picture

The closure degree at $m \cdot \ell = 21$ and beyond (64 GB host, or L2/L3); (L1, a target-free system a degree below the target system, was measured and closed on 2026-10-18); or a proof that the closure degree is $\Omega(\ell)$, for which L6 is the first lemma. Everything else is a constant, a row or bookkeeping, listed because the success criterion counts verified constants.